Day 5 afternoon — Fit, predict, check

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Monday 16 November 2026 · Pontoppidanstræde 101, room 1.001

Interpreting the fit · the mean versus one new observation · posterior predictive checks · revising the model · robust regression

Today

08:45	**Lecture A**	Regression as a generative model · scales · priors · prior predictive
09:45	**Exercise A**	Your own timing data: scales, priors, prior predictive · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Fit · mean vs new attempt · posterior predictive checks · robust
13:30	**Exercise B**	Fit, predict, check, revise the model · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

Follow-up from the morning plenary

Open points from Peer review A

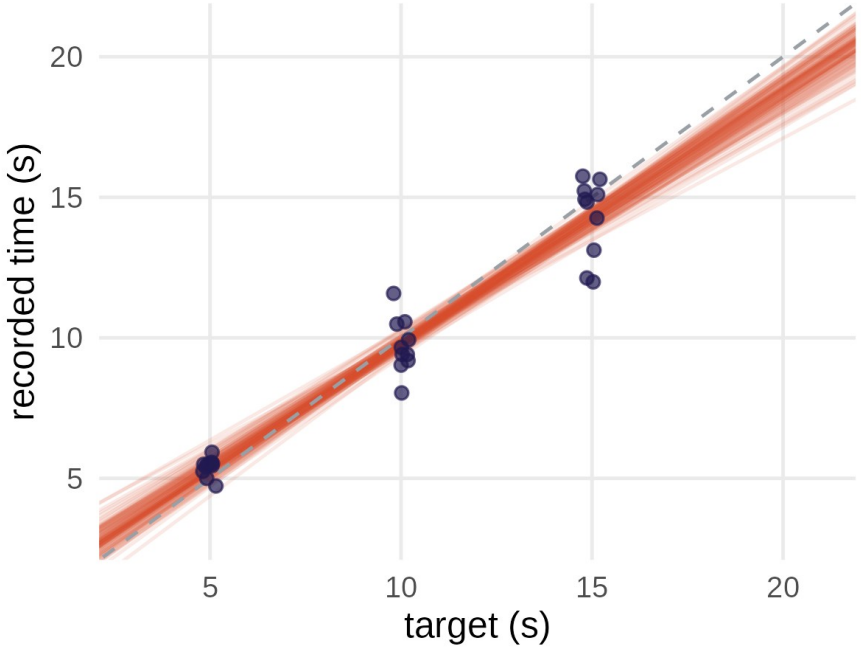
Fill in during lunch: the one or two questions the plenary did not settle.

1. …
2. …

By the end of this lecture you can

1. **Interpret** the fitted regression with probabilities about β
2. **Separate** the mean at a new x from one new observation there
3. **Check** the model with posterior predictive checks
4. **Revise** the model when a check fails, and check again
5. **Protect** the fit against an unusual observation (Student-t errors)

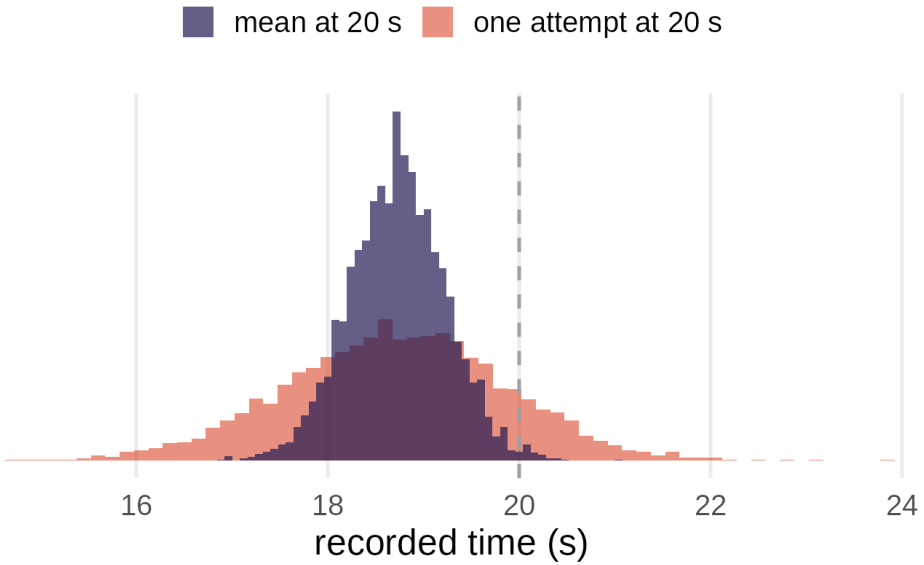
The fit



	mean	90% interval
α (s)	9.81	9.51 – 10.10
β	0.89	0.82 – 0.97
σ (s)	1.03	0.82 – 1.29

- No divergences, R-hat 1.00, ESS > 3,000.
- Each red line: $\alpha^{(s)} + \beta^{(s)} \cdot x$ for one draw.
- $P(\beta < 1 \mid y) = 0.99$: her clock very probably compresses.

At 20 s: the mean versus one attempt



Mean at 20 s:

$\mu_{20}^{(s)} = \alpha^{(s)} + \beta^{(s)} \cdot 10$

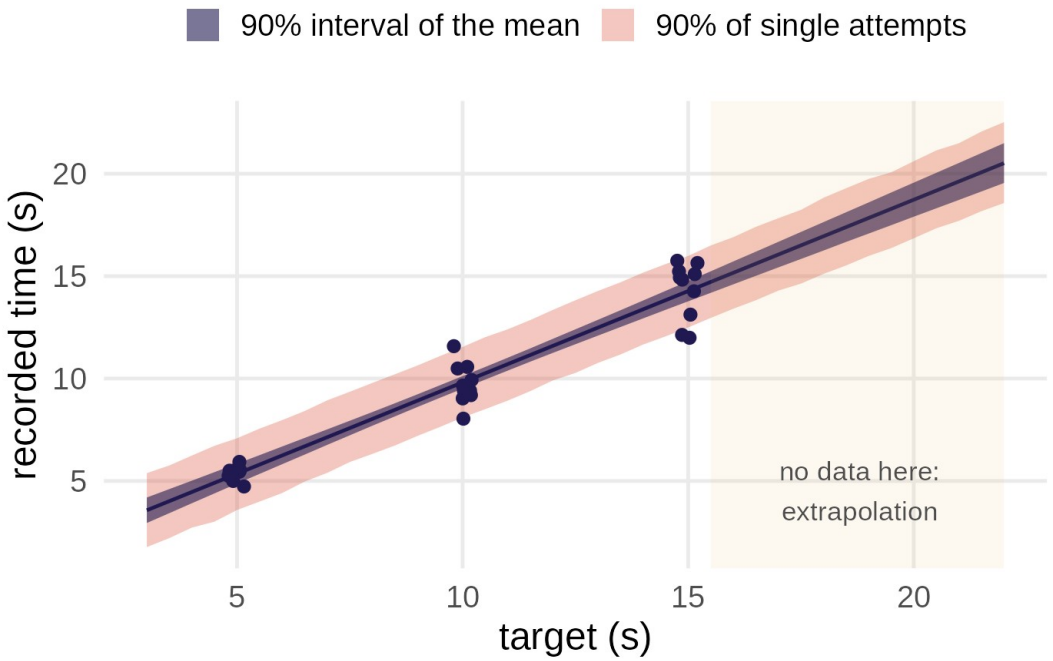
One attempt at 20 s:

$\tilde{y}_{20}^{(s)} \sim \text{Normal}(\mu_{20}^{(s)}, \sigma^{(s)})$

	90% interval	P(< 20 s)
mean	17.9 – 19.6	0.99
one attempt	16.8 – 20.5	0.87

Same idea as Day 4: an average is not one more attempt. And 20 s was never tried.

Prediction across all targets



- Dark band: where the mean line is (uncertainty about α and β).
- Light band: where 90% of single attempts fall (adds σ).
- Beyond 15 s the bands rest on the assumption that the line stays straight: extrapolation needs a reason, not just a model.

Posterior predictive checks

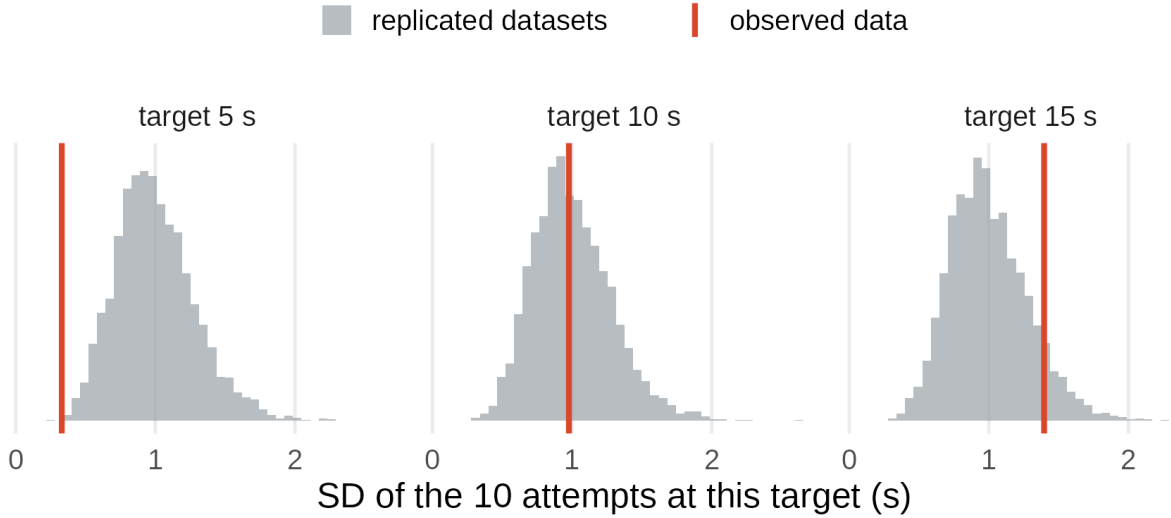
The question

If the model were right, would data like ours be typical?

1. For each draw, simulate a new dataset y_rep at the same 30 targets (generated quantities).
2. Choose a statistic T that matters for the question: here, the SD of the attempts at each target.
3. Compare T(observed data) with the distribution of T(y_rep).

```
array[N] real y_rep = normal_rng(alpha + beta * x, sigma);
```

A check: the spread at each target

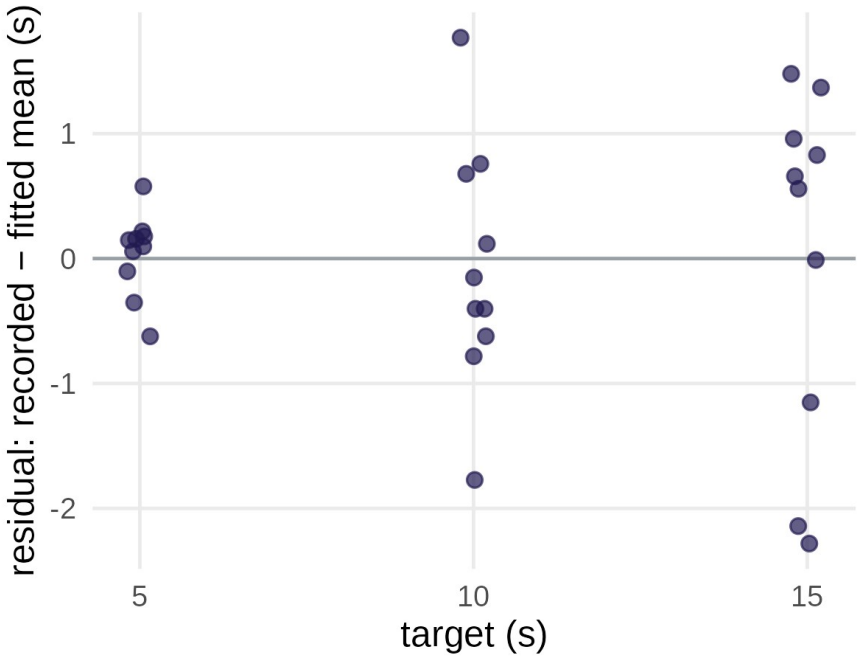


target	observed SD	replicated 90%
5 s	0.33	0.58 – 1.50
10 s	0.98	0.59 – 1.49
15 s	1.40	0.59 – 1.50

At 5 s her attempts are far less spread than any replicated dataset; at 15 s more spread than most. The model's single σ misses that the spread grows with the target.

The check found what a summary of α, β and σ cannot show.

The residuals show it too



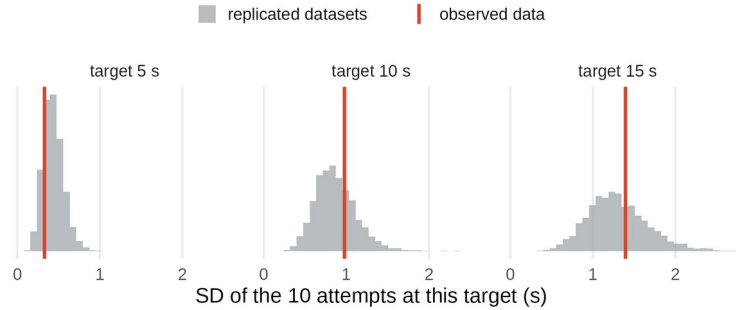
- Residual: recorded time − fitted mean.
- At 5 s within ± 0.6 s; at 15 s up to ± 2.3 s: a funnel.
- Known from timing research: the error grows with the duration (scalar timing).
- A model check should look for exactly such patterns.

Revise: the spread grows with the target

```
model {
  ...
  // spread at target t is sigma * t / 10
  y ~ normal(alpha + beta * x, sigma * (x + 10) / 10);
}
```

One changed line: σ is now the spread at a 10-s target; it halves at 5 s and doubles at 20 s.

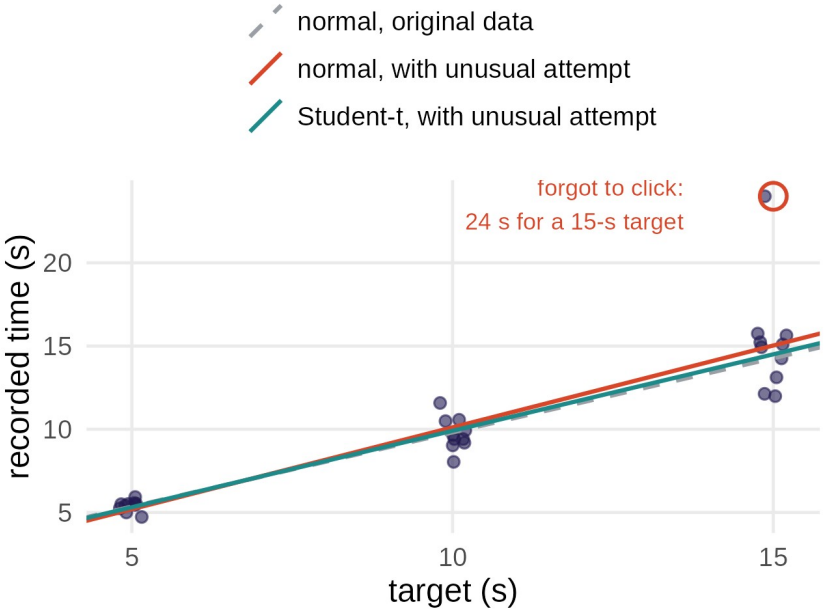
	constant σ	growing σ
β	0.89	0.88
one attempt at 20 s	16.8 – 20.5	15.5 – 21.8
SD check at 5 s	fails	passes



The same check for the revised model: the observed SDs (red) are now typical.

- The slope hardly changes.
- The prediction at 20 s becomes wider — and more honest.

An unusual attempt



- Suppose one 15-s attempt was recorded as 24 s: she forgot to click.
- Normal model: β 0.89 → 0.98, σ 1.03 → 2.08. One attempt moves both.
- Is it an error (remove it, and say so) or a real, rare event (keep it, and use a model that expects such events)?

Robust regression: Student-t errors

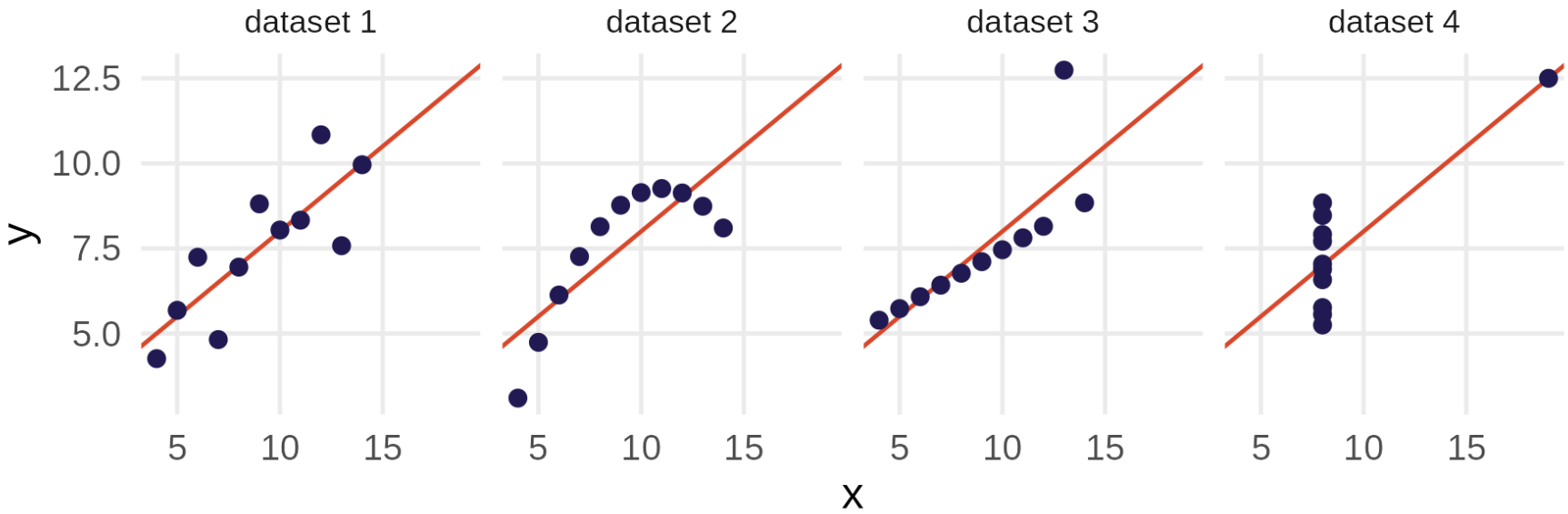
```
y ~ student_t(4, alpha + beta * x, sigma);
```

- The t distribution with 4 degrees of freedom has heavier tails: a large residual is less surprising, so one attempt pulls less.
- σ is now a scale, not the SD of the attempts.

fit	β	90% interval
normal, original data	0.89	0.82 – 0.97
normal, with 24-s attempt	0.98	0.84 – 1.13
Student-t, with 24-s attempt	0.92	0.83 – 1.00

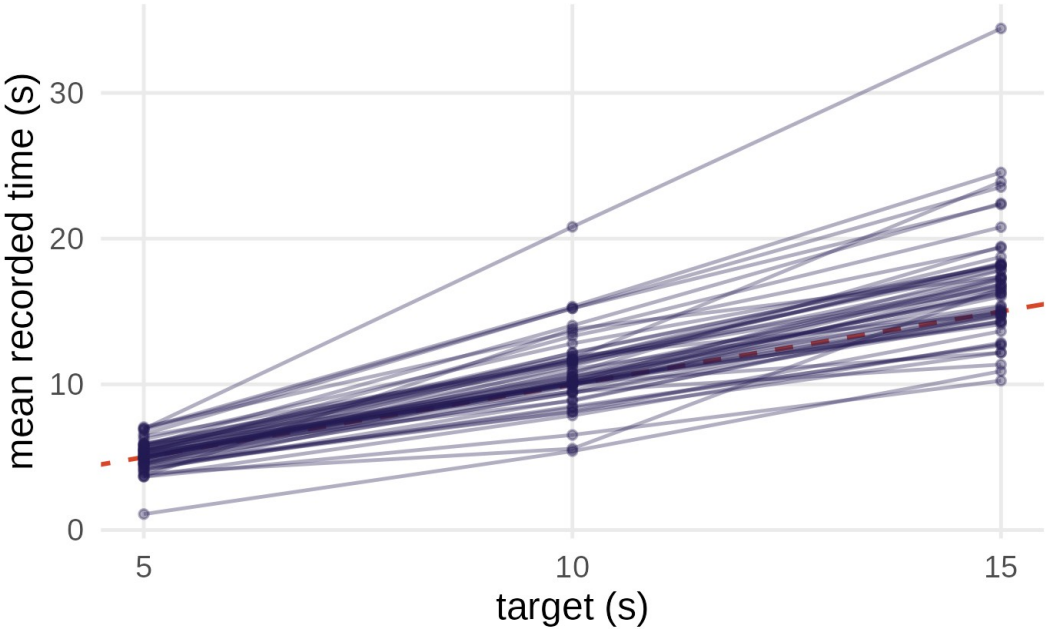
A robust model is a modelling choice, not a way to make inconvenient data disappear.

Always plot: Anscombe's quartet



All four datasets: mean of y 7.50, slope 0.50, correlation 0.82. The same regression line; four different stories.

56 clocks



one line per person (her mean at each target); dashed red: perfect timing

- The class's own Day 1 data: 56 people, ten attempts at each target without feedback.
- Every person has her own line: her own level and her own slope.
- But the lines share a common trend: longer target, longer recorded time, most of them above perfect timing.
- One regression for everybody ignores the differences; 56 separate regressions ignore what they share.

A hierarchical regression

$$y_i \sim \text{Normal}(\alpha_{j[i]} + \beta_{j[i]} \cdot x_i,\ \sigma \cdot t_i / 10)$$
$$\alpha_j \sim \text{Normal}(\mu_\alpha, \tau_\alpha), \quad \beta_j \sim \text{Normal}(\mu_\beta, \tau_\beta)$$

$j[i]$: the person of attempt i; $x_i = t_i - 10$, with t_i the target (s)

parameter	meaning
α_j, β_j	person j's own mean at 10 s and her own slope
$\mu_\alpha + \mu_\beta \cdot x$	the common trend of the class
τ_α, τ_β	how much people's levels and slopes differ
σ	spread of single attempts at a 10-s target, growing with the target (as revised this afternoon)

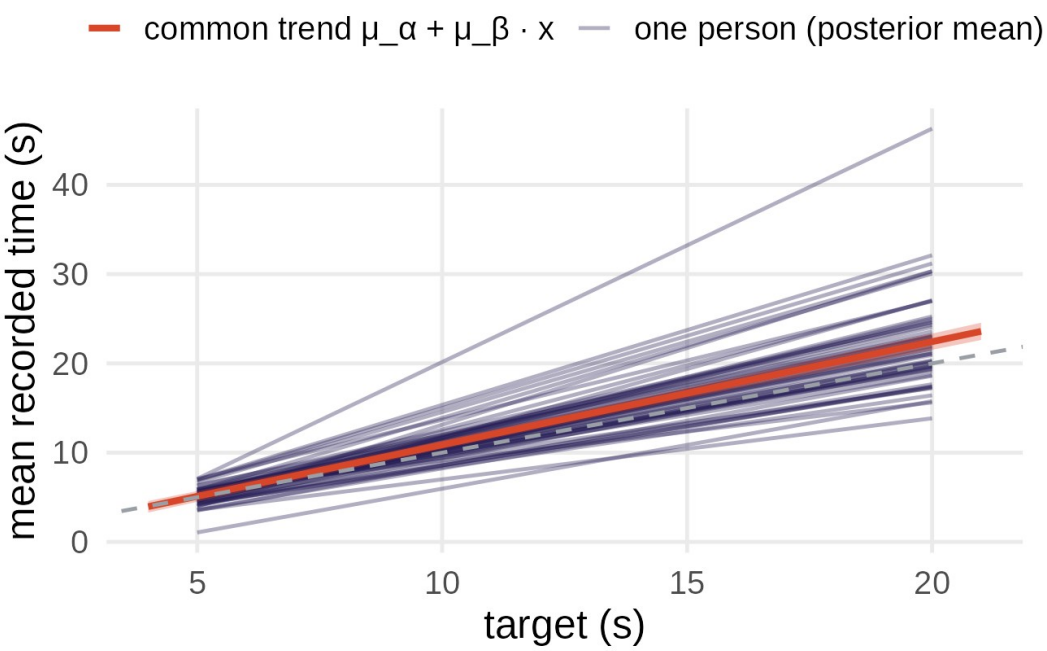
- Day 4's partial pooling, now for a line instead of a single mean.
- Each person's line is pulled towards the common trend: most for people whose own attempts say little.

The hierarchical regression in Stan

```
data {
  int<lower=1> N;                          // attempts
  int<lower=1> J;                          // people
  array[N] int<lower=1, upper=J> person; // person of attempt i: 1, ..., J
  vector[N] x;  vector[N] y;               // target - 10, recorded time
}
parameters {
  real mu_alpha;  real mu_beta;
  real<lower=0> tau_alpha;  real<lower=0> tau_beta;
  vector[J] z_alpha;  vector[J] z_beta;
  real<lower=0> sigma;
}
transformed parameters {
  vector[J] alpha = mu_alpha + tau_alpha * z_alpha;   // non-centred
  vector[J] beta  = mu_beta + tau_beta * z_beta;
}
model {
  mu_alpha ~ normal(10, 2);   mu_beta ~ normal(1, 0.5);
  tau_alpha ~ normal(0, 2);   tau_beta ~ normal(0, 0.5);
  z_alpha ~ std_normal();     z_beta ~ std_normal();
  sigma ~ normal(0, 2);
  y ~ normal(alpha[person] + beta[person] .* x,
             sigma * (x + 10) / 10);
}
```

- J and person come from R: person = match(participant, people), a number 1, …, J for every attempt.
- alpha[person] is a vector of length N: element i is alpha[person[i]]. Stan indexes with the whole array at once, so no loop is needed — as theta[person] on Day 4.
- The same as: for (i in 1:N) y[i] ~ normal(alpha[person[i]] + beta[person[i]] * x[i], …);
- .* multiplies element by element.

The common trend and the individual lines

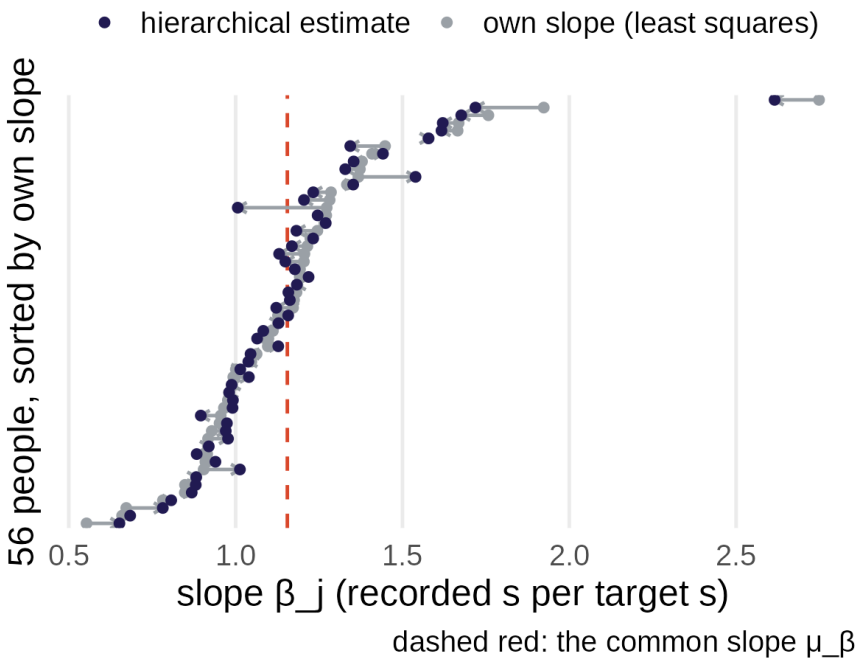


band: 90% interval of the common trend; dashed grey: perfect timing

	mean	90%
μ_α (s)	10.9	10.4 – 11.4
μ_β	1.15	1.08 – 1.23
τ_α (s)	2.24	1.90 – 2.64
τ_β	0.33	0.28 – 0.39
σ (s)	1.92	1.86 – 1.97

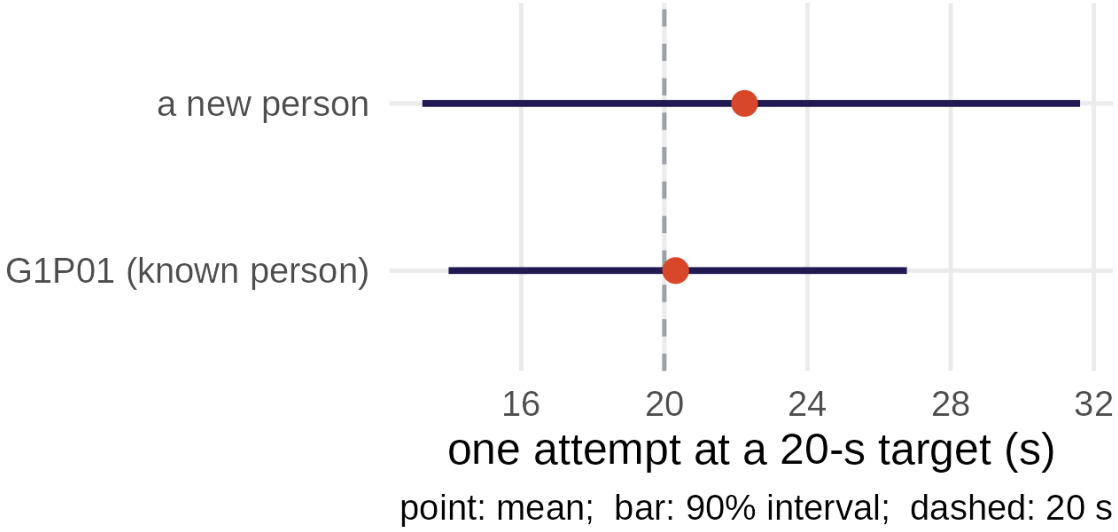
- No divergences, R-hat 1.00.
- P(μ_β > 1 | y) = 0.999: on average the class stretches long targets.
- Slopes differ: τ_β = 0.33.

Shrinkage of the slopes



- Grey: each person's own least-squares slope. Navy: the hierarchical estimate.
- Most people move very little: 30 attempts pin down a slope well.
- The largest moves are for people with erratic attempts: G5P03 (SD 7 s around her own line) moves from 1.27 to 1.01.
- The more noise in your own data, the more you borrow from the class — as on Day 4.

At 20 s: a known person versus a new person

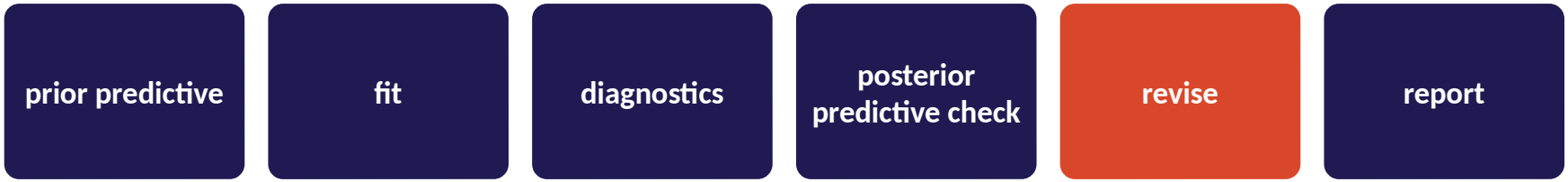


- Known person: her own α_j and β_j, plus the spread of one attempt.
- New person: draw α and β from the population first — wider.

Look closer

Fitted to G1P01's attempts alone, the same growing-spread model gives 16.8 – 23.7 s. Here 14.0 – 26.8 s. Why? σ is shared: the erratic attempts of a few people inflate it for everybody. The next revision: a σ_j per person.

The modelling workflow



- Prior predictive: does the model expect plausible data before seeing any? (morning)
- Diagnostics: did we compute the posterior correctly? (Day 3)
- Posterior predictive check: does the fitted model reproduce what matters in the data?
- Revise and check again; report the model you used and why.

Worked example: checks and predictions

Question 1

Why is the 90% interval for one attempt at 20 s wider under the growing-spread model (15.5–21.8) than under constant spread (16.8–20.5)?

Question 2

At 10 s the observed SD (0.98) lies in the middle of the replicated values. Does that prove the constant-spread model is right?

Question 3

Another person's data contain one 24-s attempt at the 15-s target. List two things you would do before choosing between removing it and a Student-t model.

Worked example: answers

Question 1

Under growing spread, σ at 20 s is twice σ at 10 s (about 1.8 s instead of 1.0 s). The data show that the spread grows, so the wider interval is the honest one; the constant model borrows the average spread from all targets.

Question 2

No. A check can only show a misfit in the statistic it looks at. At 10 s the average spread happens to fit; at 5 and 15 s it does not. Choose statistics that matter for the question, and look at several.

Question 3

Check the record (was the session interrupted? is it a typo?) and ask the person if possible; refit with and without it and report whether conclusions change. Never delete it silently.

Exercise B · 13:30 – 15:00

- Fit the regression in Stan: P(β < 1); the mean versus one attempt at 20 s.
- Posterior predictive check of the spread at each target; revise and check again.
- Challenge, no template: the hierarchical regression for the whole class.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

15:00

peer review B

If Stan does not compile on your laptop: continue on UCloud or a group member’s laptop. New roles this block.

Peer review B · 15:00 – 15:45

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

Points that need longer go to tomorrow's follow-up.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

Daily diary · 15:45 – 16:00

menti.com · code: ____ ____

1. Three things I have learned today
2. Three things I am not yet very familiar with